ALGEBRA 1 · LINEAR INEQUALITIES

Inequality Number Line

Explore & Reflect

Name _____ Date _____ Period _____

Open **Inequality Number Line** on Mr. K's Classroom: Algebra 1 → Linear Inequalities → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Is 2 a solution of $x > 2$? Is it a solution of $x \geq 2$?

2. $4 > 1$ is true. If you multiply both sides by -2, is $-8 > -2$ true?

2 Explore use the tool and record what you see

Graph and test

Set each inequality, then find one test point that makes it true.

Inequality	Open or closed circle?	Shade left or right?	A test point that works
$x > 3$			
$x \leq -1$			
$x \geq 0$			
$x < -2$			

Why does the symbol flip?

Use the second tab. Start with a true statement and do the same thing to both sides.

Start (true)	Do to both sides	Result	Still true?
$4 > 1$	add 3		
$4 > 1$	multiply by 2		
$4 > 1$	multiply by -2		
$-3 < 2$	divide by -1		
$4 > 1$	multiply by -2, then flip		

3 Write the rule fill in each blank from what you observed

1. Use an open circle for _____ and _____. Use a closed circle for _____ and _____.

2. Adding or subtracting the same number on both sides _____ the inequality symbol.

3. Multiplying or dividing both sides by a positive number _____ the symbol.

4. Multiplying or dividing both sides by a negative number _____ the symbol, because it reverses the order of numbers on the number line.

4 Check no tool: show your work

1. Solve $-3x > 12$.

2. Solve $\frac{x}{-2} \leq 5$.

3. Solve $2x - 5 \geq 7$ and describe the graph.

4. Solve $4 - x < 1$.

5 Reflect

Explain to a friend why $-2x < 6$ becomes $x > -3$, not $x < -3$.